

HW 4 Key

1. $V = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 1\}$

Zero Vector check

Is $(x, y, z) = (0, 0, 0)$
 $x=0 \quad y=0 \quad z=0$

$0+0+0 \neq 1$ V does not contain $\vec{0}$
 V is not a valid vector space

2.

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_3 = R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \end{array} \right]$$

$$\xrightarrow{R_3 = R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -2 & -4 \end{array} \right] \xrightarrow{R_3 = -\frac{1}{2}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\begin{array}{l} R_1 = R_1 - R_3 \\ R_2 = R_2 - R_3 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \begin{array}{l} \text{Cords} \\ \text{respect to } B \end{array} \text{ of vector } (2, 3, 1) \text{ w.r.t. } \boxed{(0, 1, 2)}$$

3. $v_1 = (1, 2, 1)$ $v_2 = (2, 1, 3)$ $w = (3, 3, 4)$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 1 & 3 & 4 & 4 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & -3 & -3 & -3 \\ 0 & 1 & 1 & 1 \end{array} \right] \xrightarrow{R_2 = -\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{array} \right]$$

$$\xrightarrow{\begin{array}{l} R_1 = R_1 - 2R_2 \\ R_3 = R_3 - R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Yes $w = (3, 3, 4)$ is in $\text{Span}\{v_1, v_2\}$

4. lin indpt if $c_1v_1 + c_2v_2 + c_3v_3 = 0$ when $c_1 = c_2 = c_3 = 0$

$$v_1 + v_2 = v_3 = (1, 1, 0) + (0, 1, 1) = (1, 2, 1)$$

$\begin{matrix} v_1 & & v_2 & & v_3 \end{matrix}$

NOT linearly independent

5. Check for closure under scalar multiplication

$$c(x+y) = cx + cy$$

$$c\left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}\right) = c\left(\begin{bmatrix} x_1 + x_2 + 1 \\ y_1 + y_2 \end{bmatrix}\right)$$

$$\downarrow \qquad \qquad \qquad \downarrow$$
$$c\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + c\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \qquad \qquad \qquad \begin{bmatrix} cx_1 + cx_2 + c \\ cy_1 + cy_2 \end{bmatrix}$$

$$\downarrow$$
$$\begin{bmatrix} cx_1 + cx_2 + 1 \\ cy_1 + cy_2 \end{bmatrix}$$

$\neq$

LHS $\neq$ RHS

NOT a valid Vector Space
Fails with scalar multiplication